Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 7 is a velocity / time graph for a van braking from an initial velocity of 25 m / s.

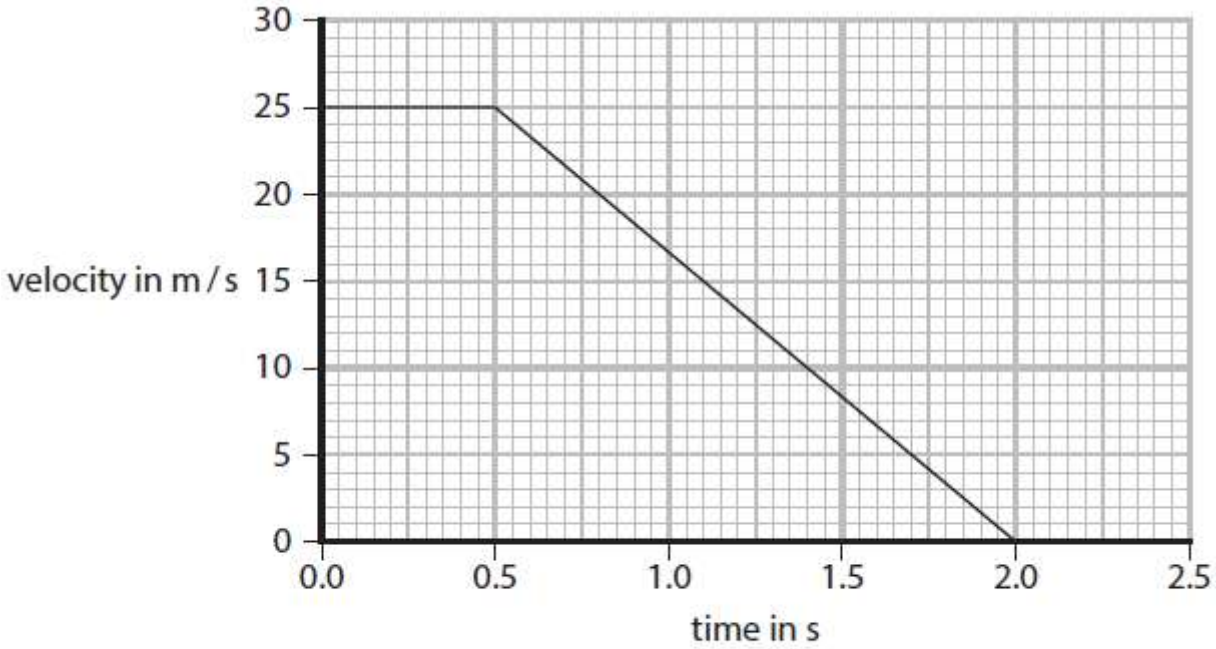


Figure 7

(i) On Figure 7 draw a similar graph for the same van braking from an initial velocity of 15 m / s.

Assume the thinking time and the braking force remain the same.

(2)

(ii) The van's brakes are applied to stop a van in a distance of 21.5 m.

The work done by the brakes is 375 kJ.
Calculate the braking force in newtons.
Use the equation

$$\text{force} = \frac{\text{work done}}{\text{distance travelled}}$$

(3)

braking force = ... N

(iii) The van is travelling along a horizontal road when the brakes are applied.

Which of these will have the same value as the work done by the brakes?

(1)

- ☐ **A** the kinetic energy of the van before braking
- ☐ **B** the kinetic energy of the van after braking
- ☐ **C** the gravitational potential energy of the van before braking
- ☐ **D** the gravitational potential energy of the van after braking

(Total for question = 6 marks)

Q2.

Figure 8 shows apparatus set up to investigate the speed of a falling object.

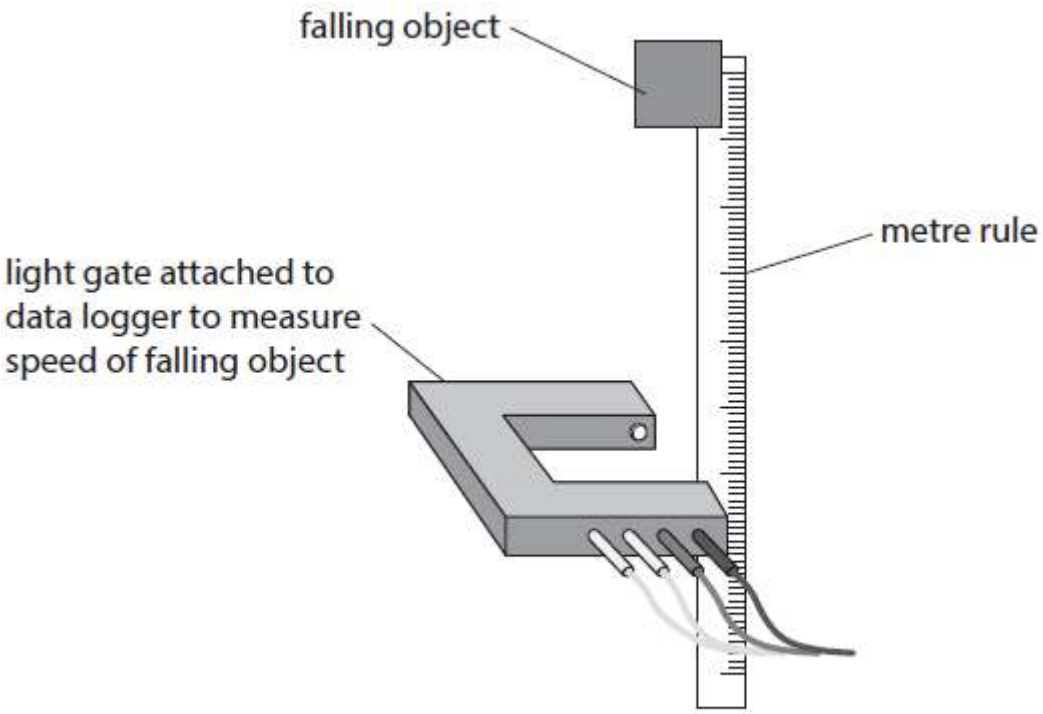


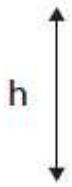
Figure 8

The object is dropped from rest at the position shown.
The object falls through the light gate.

(i) The speed of the object, after it has fallen through a height, h, is measured by the light gate.

Label Figure 8 to show the height, h.
Your label should look like this

(1)



(ii) The speed is measured at least four times for each drop height.

Give **one** reason for repeating the measurements.

(1)

.....

.....

(iii) The metre rule shown in Figure 8 is used to measure the height, h.

Suggest **two** ways to make the measurement of h as accurate as possible.

(2)

1 .....

.....

2 .....

.....

(Total for question = 4 marks)

Q3.

Some students are investigating forces.

They use a newton meter to keep a trolley stationary on a sloping runway, as shown in Figure 14.

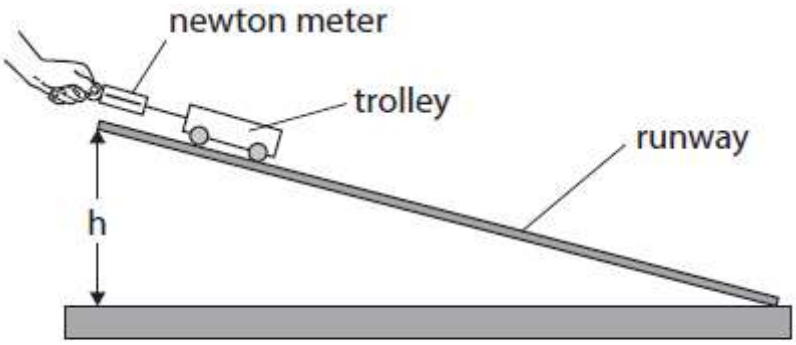


Figure 14

They measure the force needed to keep the trolley stationary on the sloping runway.

(i) They do this at four different places on the runway.

When the height, h, of the runway is 0.60 m, their four readings for force are

2.1 N 2.7 N 2.5 N 2.2 N

Calculate the mean value of these forces.

(1)

mean force = ... N

The students then measure values of the force, F, for several different values of h, the height of the runway.

Figure 15 is a graph of F against h.

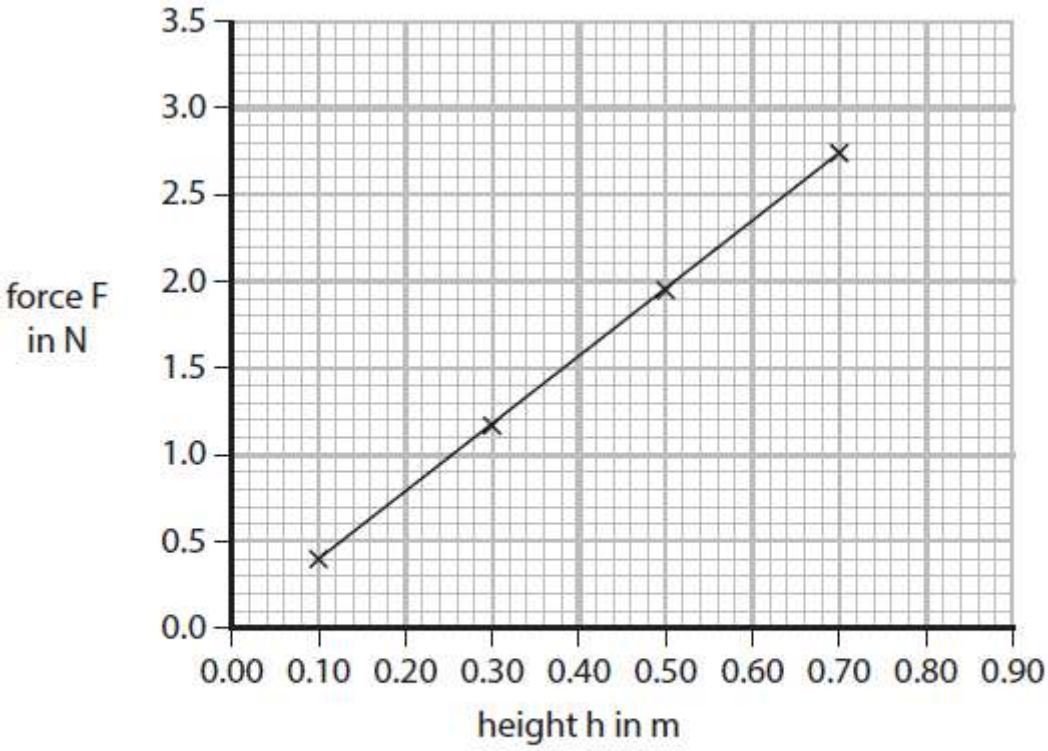


Figure 15

(ii) Use the graph in Figure 15 to determine the value of F when h = 0.80 m.

(2)

F = ... N

(iii) Determine the gradient of the graph in Figure 15.

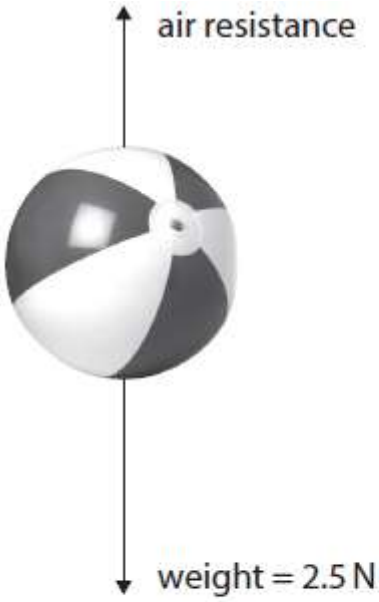
(2)

gradient = ... N/m

(Total for question = 5 marks)

Q4.

Figure 13 shows the two forces acting on a beach ball as it falls through the air.



(Source: © Stepan Bormotov/Shutterstock)

Figure 13

The resultant of these two forces is 0.92 N downwards.

(i) The mass of the beach ball is 0.25 kg.

Calculate the acceleration of the beach ball as it falls.
Use the equation

$$F = m \times a$$

(3)

acceleration = ... m/s^2

(ii) Calculate the size of the air resistance force on the beach ball.

(1)

air resistance = ... N

(Total for question = 4 marks)

Q5.

A student has some cupcake cases.

One cupcake case is shown in Figure 14.

(Source: © Anton Starikov/Shutterstock)

Figure 14

The student drops a stack of cupcake cases with the base facing downwards, as shown in Figure 15.

(Source: © Elena Schweitzer/Shutterstock)

Figure 15

The speed of the falling stack of cupcake cases depends on the number of cupcake cases in the stack.

(i) The student also has a stop clock and a metre rule.

Describe an investigation to show how the speed of the falling stack of cupcake cases depends on the number of cupcake cases in the stack.

(4)

.....

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(ii) A stack of cupcake cases has a mass of 0.005 kg.

Calculate the weight, in newtons, of the stack of cupcake cases.
Gravitational field strength = 10 N / kg

(2)

Use the equation

$$W = mg$$

weight = ... N

Figure 16 shows a cupcake case that is falling at a constant velocity.

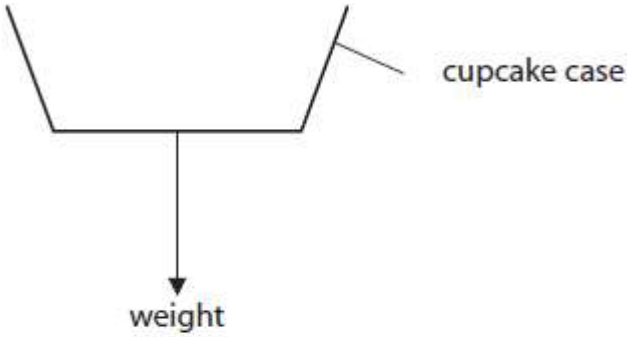


Figure 16

(iii) Draw an arrow on Figure 16 to show the force due to air resistance on the cupcake case.

(1)

(iv) State the value of the acceleration of the cupcake case when it is falling at a constant velocity

(1)

.....

(Total for question = 8 marks)

Q6.

A car is travelling down a slope at 2.0 m/s.

The car accelerates for 4.0 s.

The speed of the car increases to 12 m/s.

Calculate the acceleration of the car.

Use the equation

$$a = \frac{(v - u)}{t}$$

(2)

acceleration of the car = ... m/s^2

(Total for question = 2 marks)

Q7.

Figure 5 shows a velocity / time graph for a car moving on a test track.

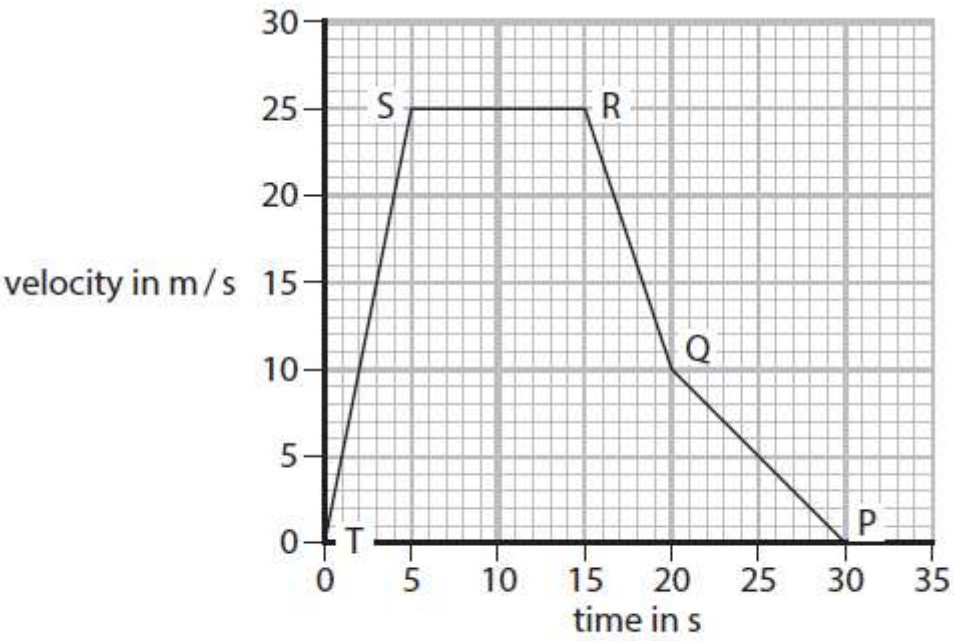


Figure 5

(i) Calculate the acceleration of the car in Figure 5 in the first 5 s.

Use the equation

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

(2)

acceleration = ... m / s^2

(ii) Calculate the distance travelled by the car in Figure 5 from 5 s to 15 s.

(2)

distance = ... m

(Total for question = 4 marks)

Q8.

A cyclist travels 1200 m in a time of 80 s.

Calculate the average speed of the cyclist.

Use the equation

$$\text{average speed} = \frac{\text{distance}}{\text{time}}$$

(2)

average speed = ... m/s

(Total for question = 2 marks)

Q9.

A car travels along a straight road.

The car accelerates at 3 m / s^2 for a time of 7 s.

Calculate the change in velocity of the car.

Use the equation

change in velocity = acceleration × time taken

(2)

change in velocity = ... m/s

(Total for question = 2 marks)

Q10.

An aircraft waits at the start of a runway.
The aircraft accelerates from a speed of 0 m/s to a speed of 80 m/s.
The acceleration of the aircraft is 4 m/s^2.

Calculate the distance, *x*, travelled by the aircraft while it is accelerating.

Use the equation

$$x = \frac{v^2 - u^2}{2a}$$

(2)

x = ... m

(Total for question = 2 marks)

Mark Scheme

Q1.

Q2.

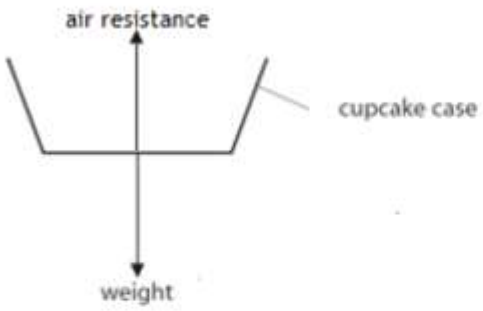
Q3.

Q4.

Q5.

Question number	Answer	Additional guidance	Mark
(i)	A description to include any 4 from: measure height (1) measure time of fall (1) use (average) speed=distance ÷ time (1) repeat with different number of cupcake cases in the stack/more cupcake cases (1) repeat **and** average time (of fall for each stack of cupcake cases) (1) plot a graph (speed of fall against number of cupcake cases dropped) (1)	 allow 'keep same height' allow in this context hold against (fixed point on) metre rule allow 'time it' accept cupcakes for cupcake cases	**(4)** **AO1**

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) (W=)0.005 x 10 evaluation (1) 0.05 (N)	 5×10^{-2} (N) do not allow power of ten error award full marks for the correct answer with no working give full credit for use of g=9.8 or 9.81 N/kg	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(iii)	 air resistance arrow (1)	judge by eye any vertical upward arrow outside or inside the cupcake case ignore length of arrow arrow need not touch cupcake holder ignore label on arrow	(1) AO2

Question number	Answer	Additional guidance	Mark
(iv)	zero / there is none / 0 / it has no acceleration	ignore 'constant' ignore units	(1) AO2

Q6.

Question number	Answer	Additional guidance	Mark
	substitution (1) (a=) $\frac{12-2(.0)}{4(.0)}$ evaluation (1) 2.5 (m/s^2)	award full marks for correct answer without working.	(2) AO2

Q7.

Q8.

Question number	Answer	Additional guidance	Mark
	substitution (1) (average speed =) $\frac{1200}{80}$ evaluation (1) 15 (m/s)	award full marks for the correct answer without working	**(2)**

Q9.

Question number	Answer	Additional guidance	Mark
	substitution (1) (change in velocity=) 3 x 7 evaluation (1) 21 (m/s)	award full marks for the correct answer with no working	**(2)** **AO2**

Q10.

Question Number	Answer	Additional guidance	Mark
	substitution (1) $\frac{80(^2)\ (-\ 0^2)}{2 \times 4}$ evaluation (1) 800 (m)	allow 1 mark for seeing $\frac{80}{8}$ ignore any minus signs award full marks for the correct answer without working	**(2)**